

Jiangxi University of Finance and Economics

Conceptual Model Report

Luckin Coffee Inc.

A Technology-Driven New Retail Coffee Enterprise

IK2015: Enterprise Architecture and Database Programming

Seminar 4

Group 4

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1 Report Basic Information

Table 1: Basic information of the conceptual modelling report.

Item	Information
Selected enterprise / business	Luckin Coffee Inc.; mobile-first coffee retail, pickup, delivery, partnership-store operation, and packaged coffee business.
Course and seminar	IK2015 Enterprise Architecture and Database Programming; Seminar 4.
Group number	Group 4.
Group members	Zhe-Kai Yang (Krust), Shi-Hang Chen (Sean), Wei-Yu Cai (Nosta), Huan-Wen Chen (Evan), Yuan-Yong Liao (Foreve).
Main deliverable	A conceptual model for the Luckin Coffee case study and a written explanation of its objects, relationships, attributes, and business meaning.

The report develops a conceptual model for Luckin Coffee by identifying key business objects, their relationships, and the variables required for measurement and decision-making. This focus fits Luckin because its operations connect mobile ordering, store fulfilment, delivery support, supply resources, and performance monitoring.

2 Modelling Scope and Design Logic

The conceptual model focuses on Luckin Coffee's daily operating logic, from customer demand to order fulfilment and performance feedback. It covers six domains: customer and membership management, digital ordering, store fulfilment, product and recipe management, supply-chain resources, and performance analytics. Together, these domains explain how Luckin converts digital demand into standardised retail service.

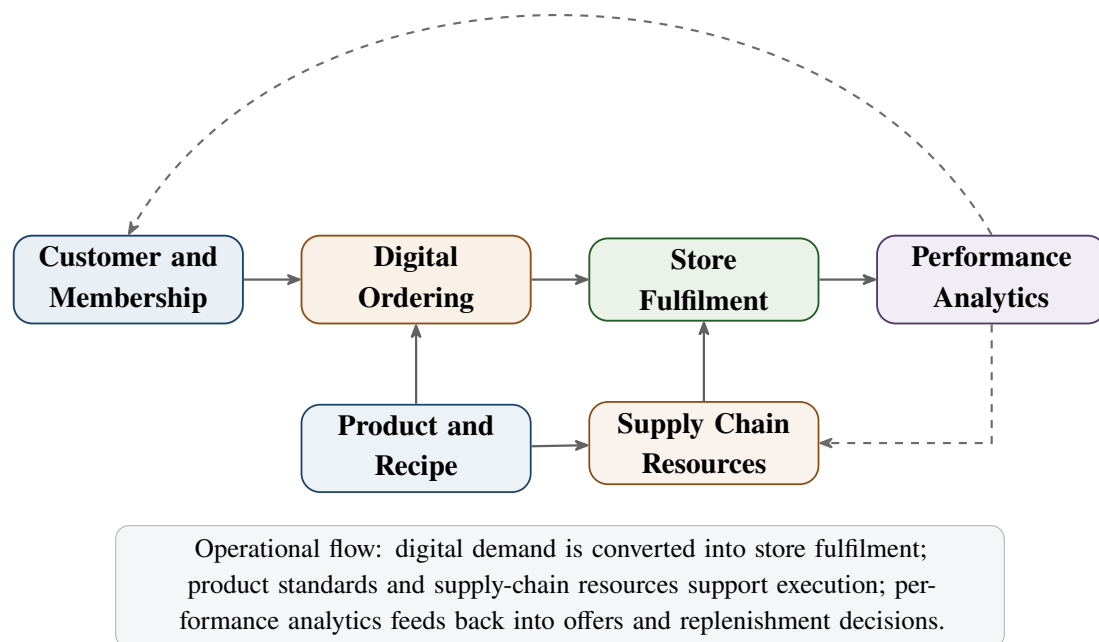


Figure 1: Scope of the conceptual model for Luckin Coffee.

The diagram uses business object boxes rather than database tables, so it remains readable for both business and technical stakeholders. The main conceptual model shows core objects, relationship IDs, and variable categories; the following tables then define representative variables and multiplicities. The relationships can later be translated into database entities, process models, or system architecture components.

3 Notation Used in the Conceptual Model

The model follows the spirit of the examples in the Seminar 4 instruction sheet. Business objects are represented as boxes, and colours separate demand, fulfilment, resource, and analytics objects. Arrows indicate reading direction, while the relationship tables specify one-to-one, one-to-many, and many-to-many multiplicities.

Table 2: Notation used in the conceptual model.

Notation	Meaning in this report
ClassVariables	Variables used to classify an object, such as customer segment, store type, order channel, payment method, or product category.
SumVariables	Quantitative values that can be summarised, such as order amount, discount amount, material cost, preparation time, or delivery fee.
AdjVariables	Context or adjustment variables used to interpret measures, such as city tier, season, channel type, queue status, or supplier lead time.
EstParameters	Estimated or derived indicators used for analysis, such as average order value, conversion rate, margin, stockout rate, or customer lifetime value.
Additional identifiers	Descriptive properties needed to identify or manage the object, such as customer ID, store ID, product name, recipe version, or supplier name.
Relationship arrows	Reading direction for the business relationship. For example, “Customer places Order” and “Order is prepared by Store”.
Multiplicity	1:N means one object can connect to many objects; M:N means many objects can connect to many objects; 1:1 means each object connects to one counterpart.

4 Conceptual Model for Luckin Coffee

LUCKIN COFFEE – Conceptual Model

LEGEND:



x Variable: indicates that the "Variable" variable has a classifying role
 Object.count – indicates that "Object" objects are counted
 Variable.sum – indicates that the "Variable" variable is summarised
 Variable.estimated – indicates that the "Variable" variable is estimated

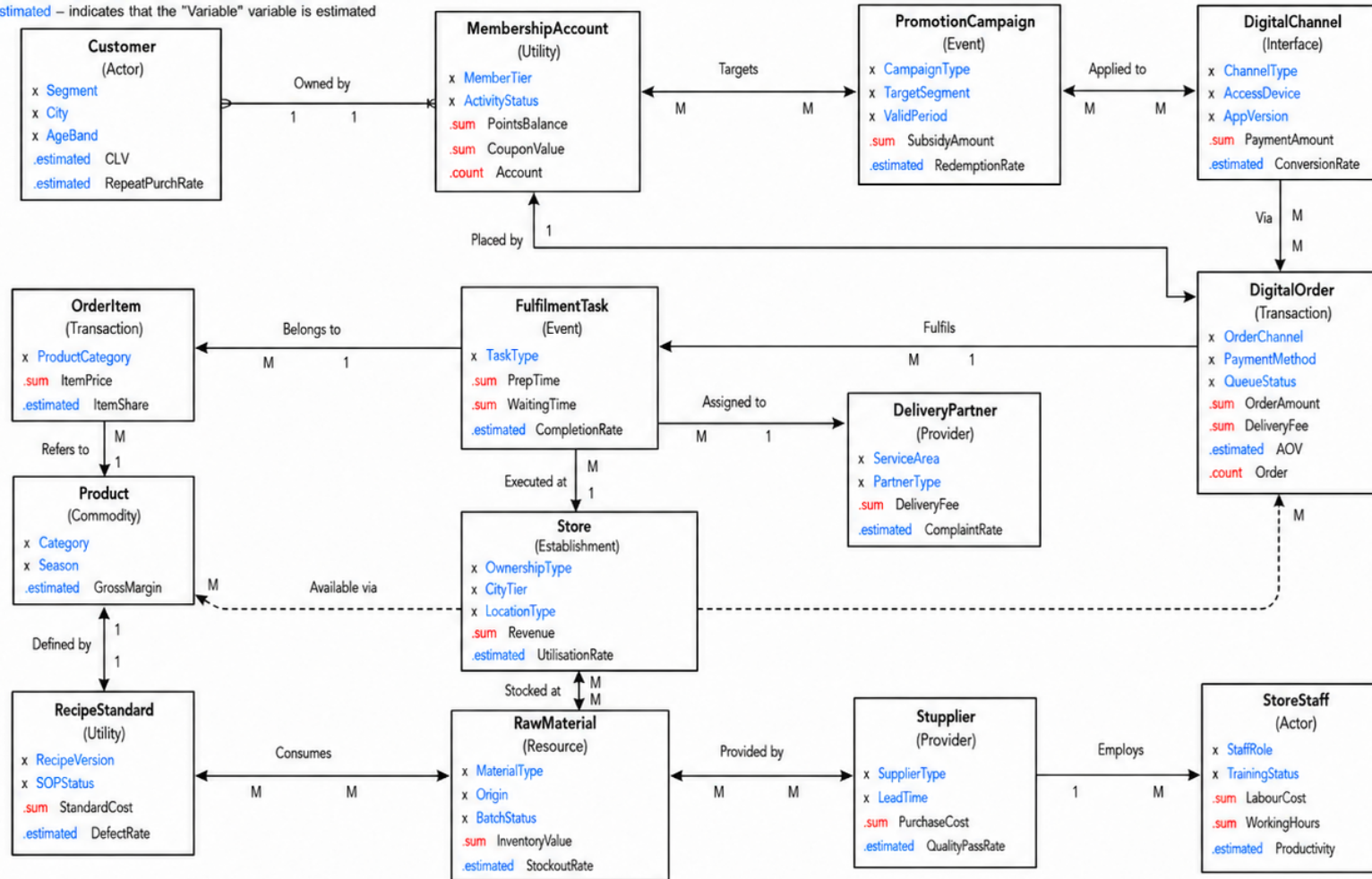


Figure 2: Conceptual model of Luckin Coffee's mobile-first retail operating system. The diagram follows the seminar notation by showing business objects, variable categories, colour-coded object types, grouped relationship identifiers, and analytical feedback links.

The diagram shows Luckin Coffee as an integrated retail platform rather than a simple chain of coffee shops. The central object is the **Digital Order**. It connects customers, membership accounts, digital channels, order items, fulfilment tasks, stores, and delivery partners. Around this core, products and recipes define what can be sold, while suppliers and materials define what can be prepared.

Table 3: Representative variables in the conceptual model.

Object	Classifying Variables (x)	Summarised Variables (.sum / .count)	Estimated Variables (.estimated)
Customer & Membership	Segment, city, age band, member tier, activity status	Points balance, coupon value, account count	CLV, repeat-purchase rate
Digital Channel	Channel type, access device, app version	Payment amount	Conversion rate
Promotion Campaign	Campaign type, target segment, valid period	Subsidy amount	Redemption rate
Digital Order & Order Item	Order channel, payment method, queue status, product category	Order amount, delivery fee, order count, item price	AOV, item share
Product & Recipe Standard	Category, season, recipe version, SOP status	Standard cost	Gross margin, defect rate
Store & Store Staff	Ownership type, city tier, location type, staff role, training status	Revenue, labour cost, working hours	Utilisation rate, productivity
Raw Material & Supplier	Material type, origin, batch status, supplier type, lead time	Inventory value, purchase cost	Stockout rate, quality-pass rate
Fulfilment & Delivery	Task type, partner type, service area	Prep time, waiting time, delivery fee	Completion rate, complaint rate

5 Explanation of Main Objects

5.1 Customer, Membership Account, and Digital Channel

The **Customer** object identifies the consumer and is classified by segment, city, and age band to explain demand differences across office workers, students, commuters, and other groups. A customer normally owns one **Membership Account**, which records digital identity, coupons, points, and order history. This reflects Luckin's dependence on digital membership rather than purely face-to-face customer management.

The **Digital Channel** object covers the mobile app, WeChat mini-program, e-commerce interfaces, and other access points. These channels capture visits, browsing, coupon redemption, and payment behaviour, making them both transaction interfaces and sources of demand data.

5.2 Promotion Campaign and Digital Order

The **Promotion Campaign** object covers coupons, membership discounts, new-product campaigns, seasonal offers, and targeted push messages. It is linked to both membership accounts and digital orders because one campaign can target many members and influence many purchases.

The **Digital Order** object is the operational centre of the model. It records order channel, payment method, queue status, order amount, delivery fee, average order value (AOV), and

order count. Separating orders from **Fulfilment Tasks** distinguishes the purchase event from the work required to complete it, such as preparation, pickup, delivery, or exception handling.

5.3 Order Item, Product, and Recipe Standard

The **Order Item** object decomposes a digital order into product lines, allowing item-level analysis of quantity, customisation, discount, popularity, and price sensitivity.

The **Product** object covers beverages, light food, packaged coffee, and merchandise. Its category, season, and gross margin help describe Luckin's frequent product testing and profitability. The **Recipe Standard** object defines the operational rule behind each product, including recipe version, SOP status, standard cost, and defect rate. This separation expresses the difference between what the customer buys and how the store prepares it.

5.4 Store, Store Staff, Raw Material, and Supplier

The **Store** object covers self-operated and partnership stores, classified by ownership type, city tier, and location type. Its revenue and utilisation rate connect digital demand with local operating capacity.

The **Store Staff** object covers employees and managers who prepare orders and follow SOPs. Staff role, training status, working hours, labour cost, and productivity are included because service consistency depends on both digital systems and human execution.

The **Raw Material** object covers coffee beans, milk, syrups, packaging, and other inputs, while the **Supplier** object identifies the external source of those materials. Together, they express the resource side of Luckin's model through origin, material type, batch status, inventory value, stockout rate, supplier type, lead time, purchase cost, and quality-pass rate.

5.5 Fulfilment Task and Delivery Partner

The **Fulfilment Task** object records the operational work created by an order, including drink preparation, pickup handling, delivery handover, and exception processing. Task type, preparation time, waiting time, and completion rate make store bottlenecks visible.

The **Delivery Partner** object covers external or platform-based logistics partners. Partner type, service area, delivery fee, and complaint rate are separated from store variables because delivery performance is partly outside direct store control but still affects customer experience.

6 Relationship Analysis

The main relationships in the model can be grouped into two layers. The first layer is the demand layer: customers use digital channels, maintain membership accounts, receive promotions, and place digital orders. The second layer is the fulfilment layer: orders contain order items, order items refer to products, products are prepared according to recipe standards, and fulfilment tasks are executed by stores and delivery partners.

Table 4: Demand, order, and fulfilment relationships in the conceptual model.

ID	Relationship	Type	Business meaning and measures enabled
D1	Digital Channel – Customer	1:N	Channels such as the app and mini-program capture customer visits, sessions, and campaign exposure.
D2	Customer – Membership Account	1:1	Customer identity is connected with points, coupons, and order history; this supports active-member and repeat-purchase analysis.
D3	Membership Account – Digital Order	1:N	One member may place many orders across time, stores, and channels; this enables order-frequency, AOV, and customer-lifetime-value analysis.
D4	Membership Account – Promotion Campaign	M:N	Campaigns target many members, while members may receive many campaigns; this supports targeting and redemption analysis.
D5	Promotion Campaign – Digital Order	M:N	Campaigns influence orders, and one order may use more than one promotional mechanism; this enables subsidy and conversion-rate tracking.
F1	Digital Order – Order Item	1:N	Order-level payment and fulfilment status are separated from product-level quantity, customisation, and item discount information.
F2	Digital Order – Fulfilment Task	1:N	A customer order can create preparation, pickup, delivery, or exception-handling tasks; this supports preparation-time and completion-rate monitoring.
F3	Fulfilment Task – Store	N:1	Many fulfilment tasks are prepared by one store, making store capacity, labour utilisation, and queue delay measurable.
F4	Store – Store Staff	1:N	Stores employ staff whose training status and productivity affect service consistency and operating efficiency.
F5	Fulfilment Task – Delivery Partner	N:1	Delivery tasks are assigned to logistics partners, allowing delivery time, delivery fee, and completion rate to be evaluated.

Table 5: Product, resource, and analytical feedback relationships in the conceptual model.

ID	Relationship	Type	Business meaning and measures enabled
S1	Order Item – Product	N:1	Many order lines can refer to the same product; this enables product sales, gross margin, and new-product success analysis.
S2	Product – Recipe Standard	1:N	A product may have recipe versions across time or regions, supporting controlled innovation and operational standardisation.
S3	Recipe Standard – Raw Material	M:N	Recipes consume multiple materials and materials may support multiple recipes; this supports ingredient-use, cost-ratio, and wastage analysis.
S4	Supplier – Raw Material	1:N	Suppliers provide material batches or categories for store replenishment; this enables lead-time, delivery-count, and quality-pass-rate evaluation.
S5	Raw Material – Store	M:N	Stores consume and hold materials from multiple batches, supporting inventory turnover, stockout, and replenishment analysis.
A1	Digital Order – Performance Result	1:N	Orders feed analytical results such as revenue, discount amount, order value, and transaction success rate.
A2	Store – Performance Result	1:N	Store operations generate periodic financial and service KPIs, including daily revenue, margin, complaints, and stockouts.
A3	Delivery Partner – Performance Result	1:N	Delivery partners generate service KPIs such as delivery completion time, delivery success rate, and logistics cost per order.
A4	Performance Result – Promotion Campaign	1:N	Analytical feedback improves future promotion design by showing which campaigns produce durable loyalty rather than only short-term discount purchases.

This relationship structure supports both operational analysis and future database design. For example, if management wants to know why delivery complaints increased in a city, the model suggests which objects should be connected: customer complaints can be traced to digital orders, fulfilment tasks, delivery partners, stores, and time periods. If management wants to evaluate a new product, the model connects order items, product categories, recipe standards, raw material consumption, promotion campaigns, and performance results.

7 Business Scenarios Supported by the Model

7.1 Scenario 1: Mobile Order and Store Pickup

A customer opens the app or WeChat mini-program, receives a personalised coupon, selects a drink, pays digitally, and chooses pickup. The model traces this process from customer and membership account to promotion campaign, digital order, order item, fulfilment task, store preparation, pickup, feedback, and KPI update.

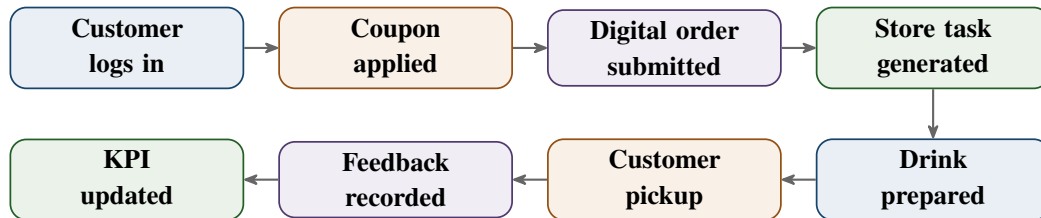


Figure 3: Pickup-order scenario supported by the conceptual model.

7.2 Scenario 2: New Product Launch and Performance Review

Luckin frequently launches new drinks and evaluates them through sales data. In the model, a new product receives a product record and recipe standard, appears in digital channels, may be supported by a campaign, and is then measured through order-item data, raw-material use, margin, complaints, and supply stability.

7.3 Scenario 3: Store Capacity and Delivery Reliability

During peak demand, the model captures pressure through digital orders and fulfilment tasks. If preparation time, waiting time, or delivery completion deteriorates, management can compare staff scheduling, material availability, delivery partner capacity, and store-level order volume.

8 Implications for Future Database Design

Although this report is conceptual rather than technical, the model provides a foundation for database design. Each object can become a candidate entity or table, and each relationship can become a foreign-key relationship, bridge table, or transaction link. For example, the many-to-many relationship between recipe standards and raw materials would require a recipe-material bridge, while the one-to-many relationship between digital orders and order items would normally become separate order and order-line tables.

Table 6: How the conceptual model can guide future database design.

Conceptual modelling decision	Database design implication
Separate customer from membership account	Customer demographic and segment information can be managed separately from app account, points, coupon, and authentication-related data.
Separate digital order from order item	Order-level payment and fulfilment status can be stored independently from product-level quantity, customisation, and item discount data.
Separate product from recipe standard	Product catalogue information can remain stable while recipe versions and SOP rules change over time.
Represent recipe-material as M:N	A bridge table would be needed to record material dosage, standard cost, and recipe version for each ingredient.
Separate fulfilment task from digital order	Operational execution can be analysed even when one order creates multiple tasks, such as preparation and delivery hand-over.
Include performance result as an analytical object	KPI reporting can aggregate operational data by store, period, region, product, campaign, and delivery partner.

The model also prevents premature database design mistakes. If the system stored only orders and products, it would be difficult to analyse fulfilment delay, recipe compliance, supply-chain pressure, or campaign effectiveness. Including fulfilment tasks, recipe standards, raw materials, suppliers, promotions, and performance results preserves the business meaning needed for later enterprise architecture and database work.

9 Conclusion

This report created a conceptual model for Luckin Coffee as a technology-driven new retail coffee enterprise. The model identifies the main objects—customer, membership account, digital channel, promotion campaign, digital order, order item, product, recipe standard, raw material, supplier, store, store staff, fulfilment task, delivery partner, and performance result—and explains how they support customer ordering, product preparation, store fulfilment, supply-chain replenishment, delivery coordination, and KPI feedback.

The model connects Luckin’s strategic logic with operational reality. Customer value is created through the coordination of digital demand capture, physical fulfilment, recipe standardisation, supply support, and performance measurement. The model is therefore detailed enough to support later database design while remaining conceptual enough to clarify the business structure before technical implementation.

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